

Optimistic Online Non-stochastic Control (via FTRL)

Naram Mhaisen & George Iosifidis

Faculty of Electrical Engineering, Mathematics, and Computer Science



Objectives

- The Online Learning Framework
 - “Optimism” in Online Learning
- Relation to Control: Online Non-stochastic Control Framework (NSC)
 - Adding the optimism flavor: Optimistic Online NSC (this work)

Introduction

Online Learning

At each time step $t = 1, \dots, T$

- Commit to an action $\mathbf{u}_t$ Based on Info available until $t - 1$
- Incur a convex loss $c_t(\mathbf{u}_t)$

Objective: $\{\mathbf{u}_t\}_{t=1}^T$ that perform well under $\{c_t(\cdot)\}_{t=1}^T$

Introduction

Online Learning

+ Untrusted predictions

Optimistic
Online Learning

At each time step $t = 1, \dots, T$

- Receive an untrusted prediction of the loss $\tilde{c}_t(\cdot)$
- Commit to an action $\mathbf{u}_t$ Based on Info available until $t - 1$
- Incur a convex loss $c_t(\mathbf{u}_t)$

Objective:- $\{\mathbf{u}_t\}_{t=1}^T$ that perform well under $\{c_t(\cdot)\}_{t=1}^T$

- The performance improves with accuracy and never falls below a strong baseline.

Introduction

Connection to Control



At each time step $t = 1, \dots, T$

- Commit to an action $\mathbf{u}_t$ Based on Info available until $t - 1$
- Transition to a new state $\mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t + \mathbf{w}_t$
- Incur a convex loss $c_t(\mathbf{x}_t, \mathbf{u}_t)$

Objective: $\{\mathbf{u}_t\}_{t=1}^T$ that perform well under $\{c_t(\cdot, \cdot)\}_{t=1}^T$.

Introduction

Parsing the title “Optimistic Online NSC”



At each time step $t = 1, \dots, T$

- Receive an untrusted prediction of the loss $\tilde{c}_t(\cdot, \cdot)$
- Commit to an action $\mathbf{u}_t$ Based on Info available until $t - 1$
- Transition to a new state $\mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t + \mathbf{w}_t$
- Incur a convex loss $c_t(\mathbf{x}_t, \mathbf{u}_t)$.

Objective: - $\{\mathbf{u}_t\}_{t=1}^T$ that perform well under $\{c_t(\cdot, \cdot)\}_{t=1}^T$

- The performance improves with accuracy and never falls below a strong baseline.

Online NSC Metric

- **Policy Regret Metric:**

$$\mathcal{R}_T \doteq \sum_{t=1}^T c_t(\mathbf{x}_t, \mathbf{u}_t) - \min_{\pi \in \Pi} \sum_{t=1}^T c_t(\mathbf{x}_t(\pi), \mathbf{u}_t(\pi)),$$

Difference between:

our accumulated costs and that of **the optimal in hindsight**

(For any $\{c_t(\cdot, \cdot)\}_{t=1}^T, \{\mathbf{w}_t\}_{t=1}^T$)

- If $R_T = \mathcal{O}(T^\alpha)$, $\alpha < 1$, Then, $\frac{R_T}{T} \rightarrow 0$
- i.e., we converge to the performance of the benchmark at the same rate.

Optimistic Online NSC

Design choices & assumption:

- **Transition:** LTI, known & stable (A, B)

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + B\mathbf{u}_t + \mathbf{w}_t.$$

- Extensible to unknown (A_t, B_t)
- **Policy class:** DAC. A policy is parametrized by $M \doteq [M^{[j]}], j \in [p]$

$$\mathbf{u}_t = \sum_{j=1}^p M_t^{[j]} \mathbf{w}_{t-j}.$$

- Approximates the class of linear controllers & has competitive ratio guarantees.
- **Cost function:** Linear (universal)

$$c_t(\mathbf{x}_t, \mathbf{u}_t) = \boldsymbol{\alpha}_t^\top \mathbf{x}_t + \boldsymbol{\beta}_t^\top \mathbf{u}_t$$

- Extensible to strongly convex, exp-concave.
- The same bounds apply to strongly convex costs (but loose)

Optimistic Online NSC

DAC motivation.

$$c_t(\mathbf{x}_t, \mathbf{u}_t) = g_t(M_{1,\dots,t})$$



From the DAC class, depends on $M_{t-p,\dots,t-1}$

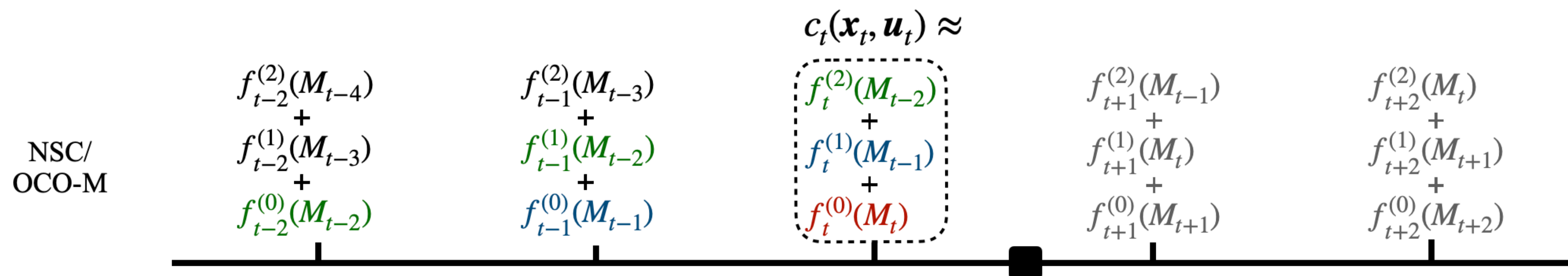
Depends on all previous actions $M_{1,\dots,t-1}$

- $\mathbf{x}_{t+1} = \sum_{i=0}^t A^i \left(B \sum_{j=1}^p (M_{t-i}^{[j]} \mathbf{w}_{t-i-j}) + \mathbf{w}_{t-i} \right)$
- $g_t(\cdot)$ is convex in the policy parameters $M_{1,\dots,t-1}$
- $g_t(\cdot)$ is mostly determined by $M_{t-p,\dots,t-1}$

Optimistic Online Control

Online Control is OCO-M

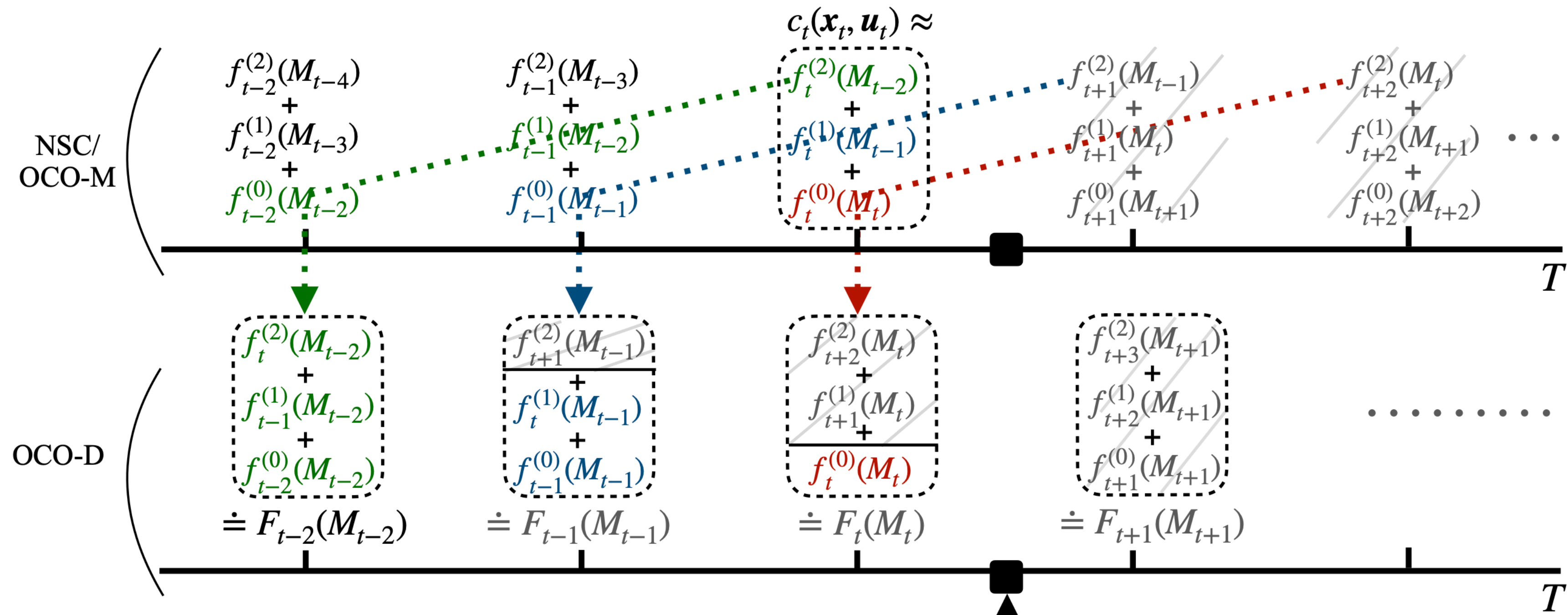
- NSC's two main ingredients:
 - Take a memory of p .
 - Solution: perform OGD on those p parameters.



$$\text{GPC: } M_{t+1} = M_t - \eta_t \nabla f_t(M_t) \longrightarrow R_T = \mathcal{O}(G\sqrt{T})$$

Optimistic Online Control

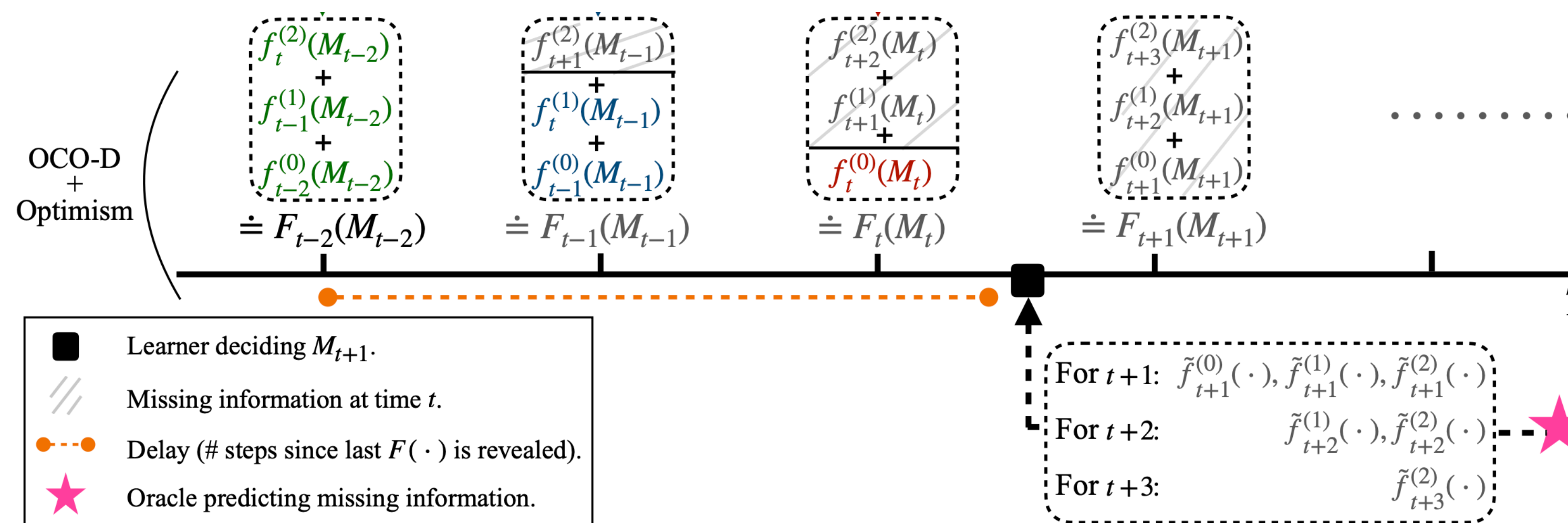
OCO-M is OCO-D (Paper idea 1)



While deciding M_{t+1} , feedback only up to F_{t-p} is available

Optimistic Online Control

OCO-D learning via Optimistic FTRL (Paper idea 2)



$$M_{t+1} = \arg \min_M \underbrace{F_{1:t-p}(M)}_{\text{Past costs}} + \underbrace{\tilde{F}_{t-p+1:t+1}(M)}_{\text{Future pred.}} + \underbrace{\frac{\lambda_{t+1}}{2} \|M\|^2}_{\text{Regs.}}$$

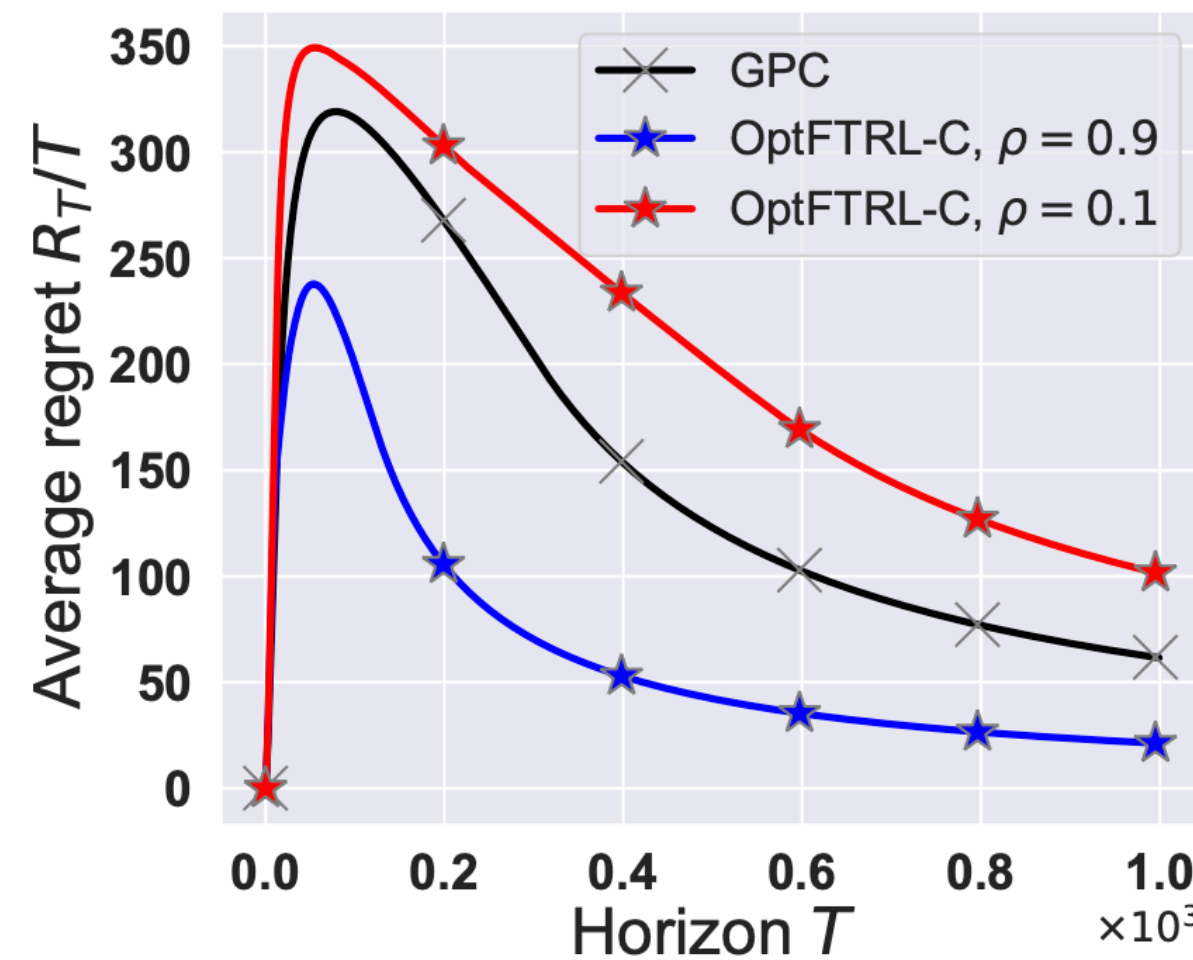
$$\lambda_{t+1} \propto \sqrt{\underbrace{\Delta_{1:t-p}}_{\text{Witnessed pred. error}}}, \quad \Delta_t = \|\nabla F_{t-p}(M_{t-p}) - \nabla \tilde{F}_{t-p}(M_{t-p})\|^2$$

$$R_T = \mathcal{O} \left(G \sqrt{\sum_t \Delta_t} \right)$$

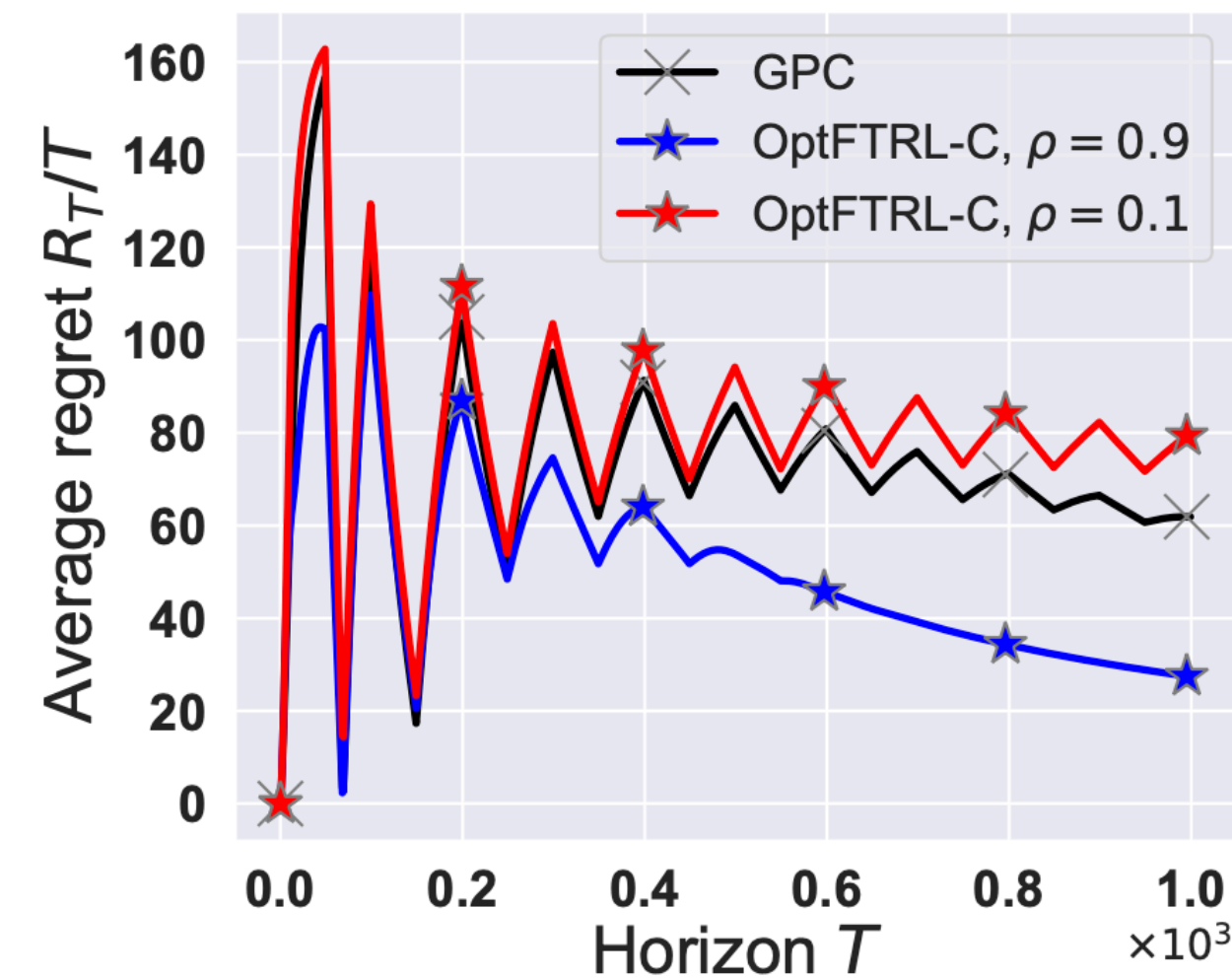
Optimistic Online Control

Numerical Example & Conclusion

$$R_T = \mathcal{O} \left(G \sqrt{\sum_t \Delta_t} \right) \begin{cases} \nearrow \mathcal{O}(1) \\ \searrow \mathcal{O}(\sqrt{T}) \end{cases}$$



(a) Stationary costs.



(b) Alternating costs.

Thanks

- More info in the paper:



- NSC Tutorial (video): <https://sites.google.com/view/nsc-tutorial/home>
- Optimistic Learning Tutorial (slides): <https://www.futurenetworkslab.net/>